

Taste and smell

Unlike other senses, taste and smell work by detecting chemicals, often in incredibly small amounts. Animals use taste to identify food and to avoid harmful substances. Mammals taste with their taste buds, which are mainly on their tongues, but other animals taste with different body parts, or even their entire body surface. Smell detects substances that spread through air or water. Some odors warn of danger, but others are attractive, helping animals to find food or a potential mate.

Tasting the air

When exploring its surroundings, a rat snake constantly flicks its tongue in and out in order to collect chemicals from the air, which it analyzes in the roof of its mouth. Many snakes have forked tongues. They may use these to find the chemical's source, just as we use two ears to locate a sound. If the chemical is stronger on one tip than the other, this may guide a snake toward prey or away from a predator.

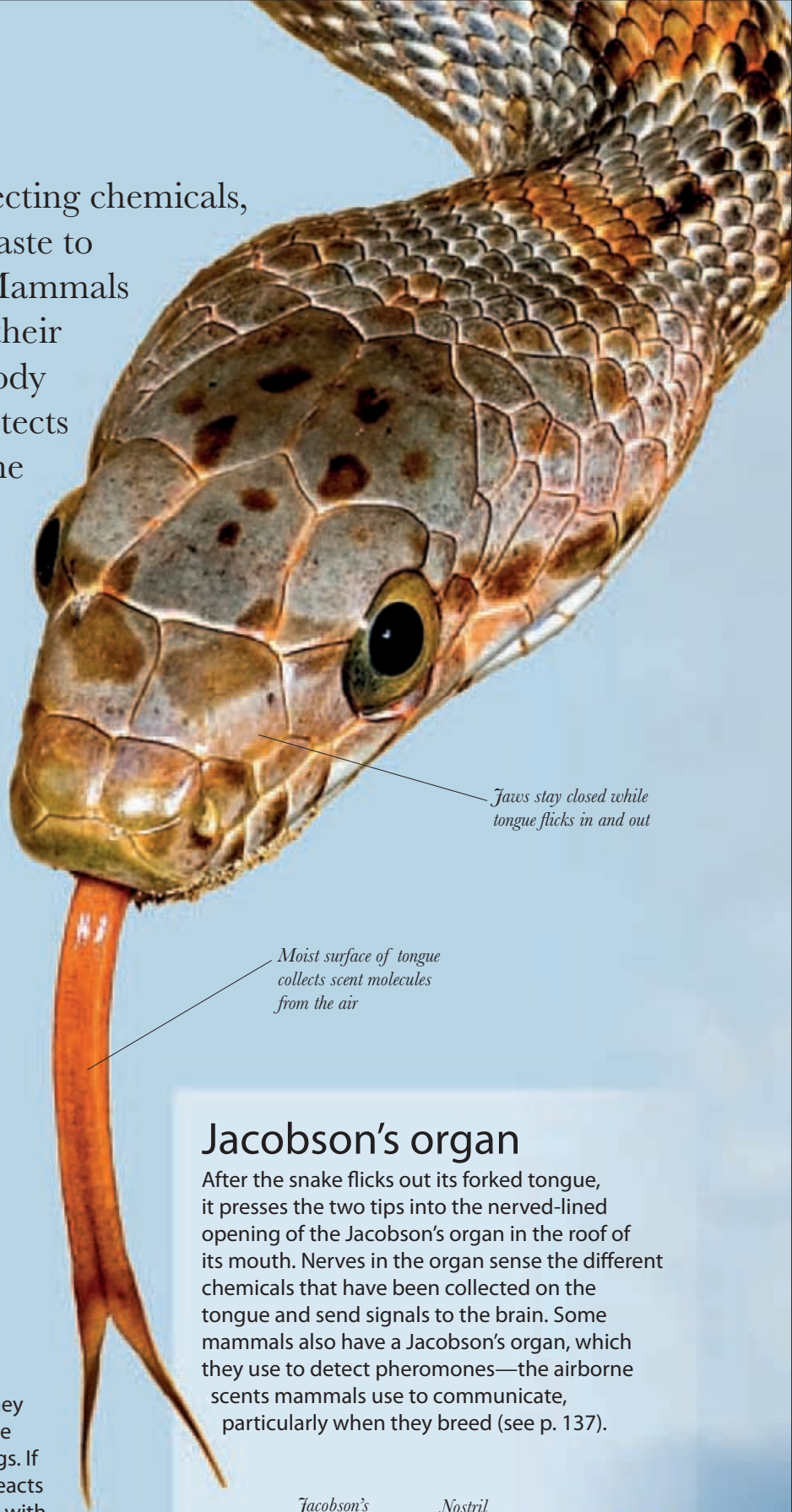


Sensitive feet

This malachite butterfly is not simply perching on a leaf—it is tasting it as well. Butterflies can do this because they have taste sensors on their feet. This talent helps females to find the right plants to lay their eggs on, and it helps adults to track down food. Butterflies are especially attracted by sweet tastes. Their feet are more than 200 times as sensitive to sugar as human tongues.

Underwater smell

Orange cup corals cannot see their prey, but they are sensitive to movement and to waterborne chemicals given off by living things. If a fish swims nearby, the coral reacts to its scent and reaches for it with its stinging tentacles. In open water, many swimming predators use similar senses to track down their prey. Sharks are extremely good at detecting amino acids from injured flesh, smelling them even when they are diluted more than 10 billion times.



Jaws stay closed while tongue flicks in and out

Moist surface of tongue collects scent molecules from the air

Jacobson's organ

After the snake flicks out its forked tongue, it presses the two tips into the nerved-lined opening of the Jacobson's organ in the roof of its mouth. Nerves in the organ sense the different chemicals that have been collected on the tongue and send signals to the brain. Some mammals also have a Jacobson's organ, which they use to detect pheromones—the airborne scents mammals use to communicate, particularly when they breed (see p. 137).

